Main datasets (fixes of AI code)

#	Paper	Paper (PDF)	Data / code to download
1	**AIDev** – Li et al. 2025/26	arXiv 2602.09185	huggingface.co/datasets/hao-li/AIDev (v3 / v4); check for v5
2	**Who Finishes the Job?** – Takerngsaksiri 2026	arxiv.2609.26847	github.com/wannita901/pr-fix-authorship (= evolution-assurance)
3	**Do These Violent Delights Have Violent Ends?** – Xia & Miller 2026	arXiv 2607.09902	github.com/post-merge-reality/post-merge-reality
4	**To What Extent Does Agent-generated Code Require Maintenance?** – Sawada 2026 (EASE)	arXiv 2605.06464	github.com/ShouSawa/AI-Code-Maintainability
5	**On the Use of Agentic Coding** – Watanabe 2025	arXiv 2509.14745	github.com/mmikuu/OnTheUseOfAgenticCoding
6	**Where Are LLMs for Code Generation on GitHub?** – Yu 2024	arXiv 2406.19544	github.com/lkpttxg/WhereCodeGenerationByLLM
7	**Humans Integrate, Agents Fix** – Khemissi 2026	arXiv.2604.04059	Figshare
8	**Why Are AI Agent–Involved Pull Requests (Fix-Related) Remain Unmerged? –** Alam et al. 2026	arXiv 2602.00164	**Zenodo:** **doi.org/10.5281/zenodo.18105922** · 8,106 AI fix PRs from AIDev-pop; *AI fix attempts before merge, not fixes of AI code*

Methods and taxonomies (for Q1, Q2, Q3)

#	Paper	Paper (PDF)	Data / code to download
9	**What Do AI Agents Actually Change?** – Dovhoshliubnyi 2026 (SSBSE)	SSBSE 2026 page	github.com/5uper6rain/ssbse-challenge-2026
10	**Code Generation Errors Made by LLMs** – Wang 2025 (ICSE)	arXiv 2406.08731	github.com/ma-labo/defects4codellm
11	**Learning from 53.6K Developer Edits of AI Code (DECODE)** – Liang 2026	arXiv 2607.25130	code: github.com/jennytliang/decode · data: Hugging Face `jtliang/decode`

Tools for Q3 (how was it fixed)

#	Tool	Download
12	**RefactoringMiner** (Tsantalis), supports Python	github.com/tsantalis/RefactoringMiner
13	Py-RefactoringMiner (Dilhara)	github.com/maldil/RefactoringMiner
14	PyRef (Atwi 2021)	github.com/PyRef/PyRef

Paper	AI code is a…	Fix is a…	Kind
Who Finishes	**PR**	**PR**	**PR → fix PR**
Khemissi	**PR**	**PR**	**PR → fix PR**
Watanabe	**PR**	**commit inside the same PR**	**PR → fix commit**
Xia & Miller	**commit**	**later commit**	**commit → fix commit**
Sawada	**commit (file)**	**later commit**	**commit → fix commit**
Yu	**file**	**later change**	**file → fix**
Alam, Dovhoshliubnyi, Wang	**—**	**—**	**labels or categories only**

Main datasets (fixes of AI code)

1. AIDev – Li et al. 2025/26

- 2.74M pull requests written by AI agents (Codex, Copilot, Devin, Cursor, Claude Code, Jules); AIDev-pop has 33,596 PRs from popular repos.
- **What we take:** the raw source of AI PRs for linker.py, to build your own dataset (newer, Python only).

2. Who Finishes the Job? – Takerngsaksiri 2026

- 2,510 AI PR → later PR pairs. An LLM judge confirmed 334 direct fixes. 80% of the fixes are written by an AI agent, and the median is 1 day after the AI PR.
- **What we take:** linker.py + the LLM judge (the first half of your pipeline), and the **323 clean PR → fix PR pairs with code**.

3. Xia & Miller 2026 – Post-merge fate of agentic code

- in 182 repos (May 2025 → May 2026), 26.9% of AI commits are fixed later vs 25.8% of human commits. The first later change is a fix for 21.3% of AI code vs 17.3% of human code.
- **What we take:**
    - 7,822 fixed AI commits (agent names mined from GitHub; 3,891 with fix code)
    - the maintenance classes for **Q1 (why)**
    - the line-tracking method

4. Sawada 2026 – How much maintenance AI code needs

- files created by AI get about as many fixes as human files (about 14% of later commits). Copilot files are mostly fixed by AI; Devin and Cursor files mostly by humans.
- **What we take:** 216 fix commits on AI-created files (all with code now), and the AI-detection rules.

5. Watanabe 2025 – Claude Code pull requests

- 75% of Claude Code PRs are merged, 36% of the merged ones needed a revision, and “fix” is the most common revision.
- **What we take:** 102 fix commits **inside** PRs (before merge), useful for comparison.

6. Yu 2024 – ChatGPT/Copilot code on GitHub

- only 8% of files with ChatGPT/Copilot code get a fix later; most changes add features.
- **What we take:** 113 files with fixes, as an older baseline (chat LLMs, before agents).

7. Khemissi 2026 – Humans Integrate, Agents Fix

- 2,010 later commits reference an AI PR. “Fix” is the top reason (in the hand-labelled sample), and 42% of human commits are co-authored by an agent.
- **What we take: 1,032 fixes of AI PRs with 937 code patches**, a second source of fixes.

8. Alam 2026 – Why AI fix PRs are not merged

    - 8,106 AI fix PRs, of which 65% are merged.

    - Rejected mostly because another PR solved it (22%), tests failed (18%), or the fix was wrong (15%).
- **What we take:**
    - the **AIDev-pop tables missing in v4** (`pr_task_type`, `pr_timeline`) for linker.py
    - related work (AI fix attempts, not fixes of AI code)

Methods and taxonomies (for Q1, Q2, Q3)

9. Dovhoshliubnyi 2026 – What AI agents change

in AI performance PRs, the most common changes are renames (23%), new objects (16%) and type changes (14%).

- **What we take:** the 18-type **code-change taxonomy and hunk classifier** for **Q3 (how)**.

10. Wang 2025 – Errors in LLM code

The most common errors are wrong logic (21%), missing steps (16%) and wrong conditions (14%). Most errors are a missing or wrong code block.

- **What we take:** the **error taxonomy** (13 + 14 types) for **Q2 (what was wrong)**.

11. DECODE – Liang 2026

- 53,614 developer edits of AI code, 75% in Python. 56% of edits change what the code does.
- **What we take:** before/after edit extraction and edit-type labels (edits before submission).

Tools for Q3 (how was it fixed)

12. RefactoringMiner (Tsantalis): it now supports Python and whole PRs. It finds the exact refactorings in each fix.

13. Py-RefactoringMiner and **14. PyRef:** backup Python tools.